MIL-L-25370D(USAF)
4 October 1966
Superseding
MIL-L-25370C(USAF)
3 December 1964

MILITARY SPECIFICATION

LIFE PRESERVER INFLATING ASSEMBLY, MA-1

1. SCOPE

1.1 This specification covers one type of life preserver inflating assembly designated MA-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

V-T-295	Thread, Nylon
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-576	Box, Wood, Cleated Veneer, Paper Overlaid
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-B-676	Boxes, Set-Up
PPP-C-843	Cushioning Material, Cellulosic
PPP-T-60	Tape, Pressure-Sensitive Adhesive Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-C-25369	Cylinder, Carbon Dioxide, for Inflation of Pneumatic Life Preservers

STANDARDS

Federal

FED-STD-151	Metals; Test Methods

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS33586	Metals, Definition of Dissimilar

DRAWINGS

Air Force

52A6598	Stem - Inflator Valve, Life Preserver Bladder
52A6600	Cap - Inflator Valve, Pneumatic Life Preserver
56K3881	Life Preserver - Underarm, Casualty, Type MB-1, Assembly of
58A3567	Gasket - Life Preserver Inflator Inlet

FSC 4220

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58A3568	Gasket - Life Preserver Inflator Stem, Upper
58A3569	Gasket - Life Preserver Inflator Stem, Lower

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Components. The inflating assembly shall consist of:

Item No.	Quantity	Item Name
1	1	Inflating assembly housing (see 3.5.1)
2	1	Lanyard (see 3.5.2)
3	3	Gaskets (see 3.5.3)
4	1	Valve Stem Cap (see 3.5.4).

3.4 Materials

3.4.1 Age. Unless otherwise specifically approved by the procuring activity all materials and component parts used in the inflating assembly shall have been manufactured not more than 12 months prior to the date of delivery of the Government.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to salt spray or any other atmospheric conditions likely to be met in storage or normal service.

3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

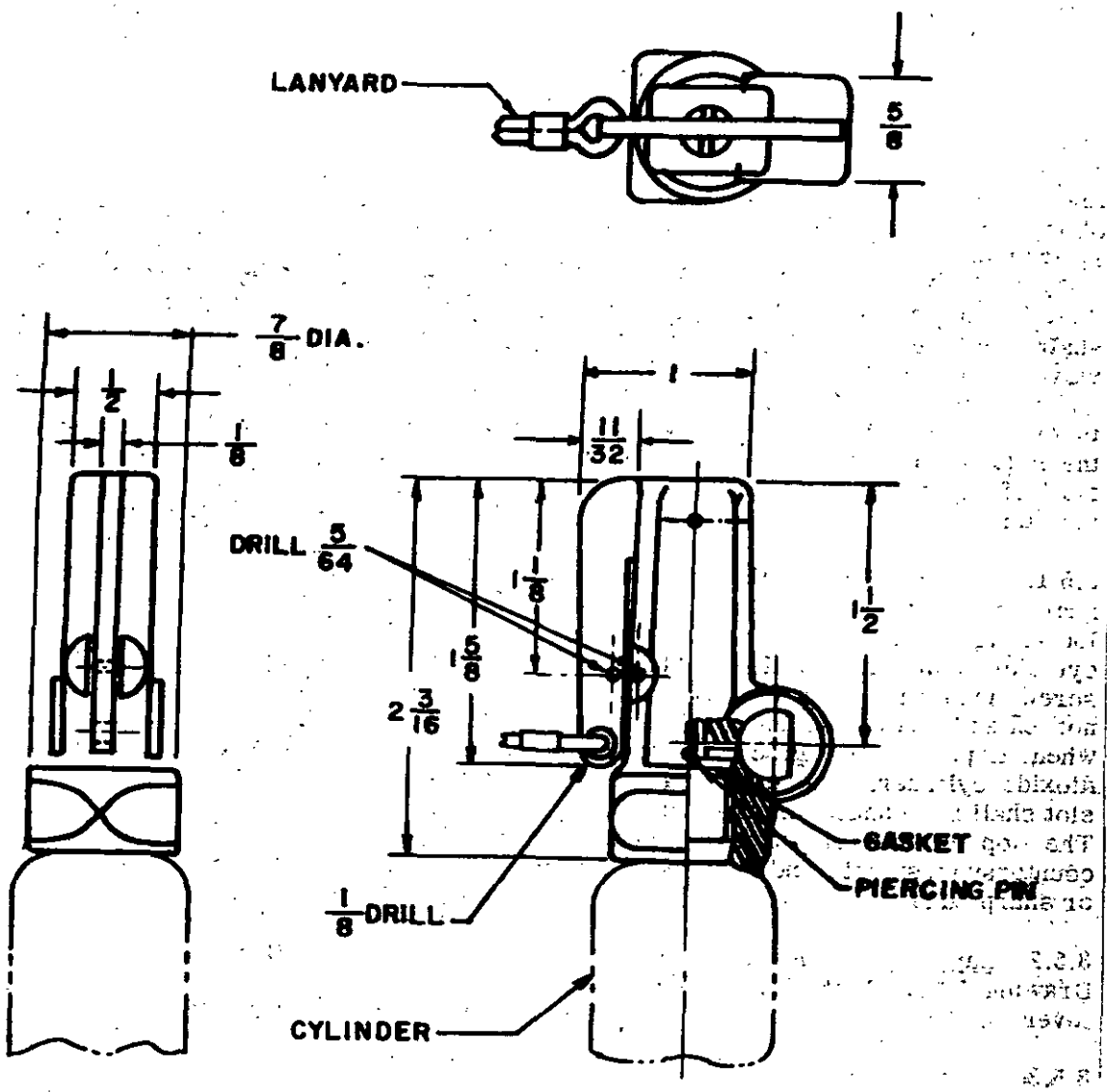
3.5 Design and construction. The dimensions and the shape of the inflating assembly shall conform to figure 1.

3.5.1 Inflating assembly housing. The inflating assembly housing shall be high pressure die-cast aluminum and shall be anodized and treated with a dichromate seal. The discharge mechanism shall consist of an external lever and an internal piercing pin and shall have a seal that will prevent leakage of life preserver cell pressure when the lever is in any position. The inlet of the inflating assembly shall be threaded to accept a carbon-dioxide cylinder conforming to MIL-C-25369. A lock shall be provided to retain the carbon-dioxide cylinder in the inlet. An outlet shall be provided to mount the inflating assembly on a life preserver cell. Passages shall be provided for conducting gas from the punctured carbon-dioxide cylinder to the outlet when the lever is actuated.

3.5.1.1 Piercing pin. The piercing pin shall be cadmium-plated, hardened steel, having a case depth of 0.010 inch. A spring shall be provided to retract the pin immediately from the opening of a punctured cylinder.

3.5.1.2 Lever. The lever shall be 1/8 inch thick; cadmium plated, hardened steel, having a case depth of 0.010 inch. The lever shall be held in position with

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UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE INCHES AND
TOLERANCES ARE ±1/32

FIGURE 1. Inflating Assembly

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the staked stainless steel pin. In the unoperated position the lever shall lay against the side of the inflating assembly as shown in figure 1. Thread conforming to type III, letter size E, color optional of V-T-295 shall be used for securing the lever in the unoperated position. All edges of the lever shall have a nominal radius of ten-thousands of an inch.

3.5.1.3 Outlet. Each inflating assembly shall have an outlet for mounting on a valve stem conforming to Drawing 52A6598 as shown in figure 2. The gas passages in the outlet shall align with the hole in the side of the stem regardless of which side of the outlet is toward the base of the stem.

3.5.1.4 Lock. Each inflating assembly shall have screw-type lock to prevent loosening of an installed carbon-dioxide cylinder conforming to MIL-C-25369. The screw head shall be a slot type and shall not extend above the inflating assembly when adjusted to tighten the carbon-dioxide cylinder. The screw head and the slot shall be rounded and free from burrs. The top of the tapped holes shall be countersunk smooth and free from burrs or sharp edges.

3.5.2 Lanyard. A lanyard conforming to Drawing 56K3881 shall be attached to the lever.

3.5.3 Gaskets. Gaskets shall be provided to seal the attachment points of the outlet and the inlet of the inflating assembly. The inlet gasket shall conform to Drawing 58A3567. The outlet gaskets shall conform to Drawings 58A3568 and 58A3569.

3.5.4 Valve stem cap. A valve stem cap conforming to Drawing 52A6600 shall be provided to hold the inflating assembly in place on the valve stem.

3.6 Performance

3.6.1 Operation test without cylinder. A load of not more than 20 pounds attached to the end of the lever arm of the inflating assembly shall operate the piercing mechanism.

3.6.2 Lever operation. The lever shall be designed so that an installed cylinder conforming to MIL-C-25369 will be punctured and the piercing pin retracted to its original position with a travel of the lever through an arc of not more than 90 degrees in line with the longitudinal axis of the inflating assembly.

3.6.3 Piercing pin operation. The point of the pin shall be positioned not less than 1/32 inch behind the surface of the inflating assembly gasket (Drawing 58A3567).

3.6.4 Hydrostatic pressure. The inflating assembly shall satisfactorily withstand a hydrostatic pressure of 1,500 pounds per square inch (psi) without deformation or leakage.

3.6.5 Salt spray. After subjection to the salt-spray test and with a load of 20 pounds on the lever arm, the inflating assembly shall pierce and discharge a cylinder conforming to MIL-C-25369 in not more than 2-1/2 seconds.

3.6.6 Leakage. With the lever moved slowly through a normal cycle of operation while the inflating assembly is submerged in water and is subjected to internal pressures of 2 psi and 40 psi (in this order) for not less than 10 seconds each, the inflating assembly shall not leak at any point.

3.6.7 Operation test with cylinder. With a load of 20 pounds on the lever arm, the inflating assembly shall pierce and

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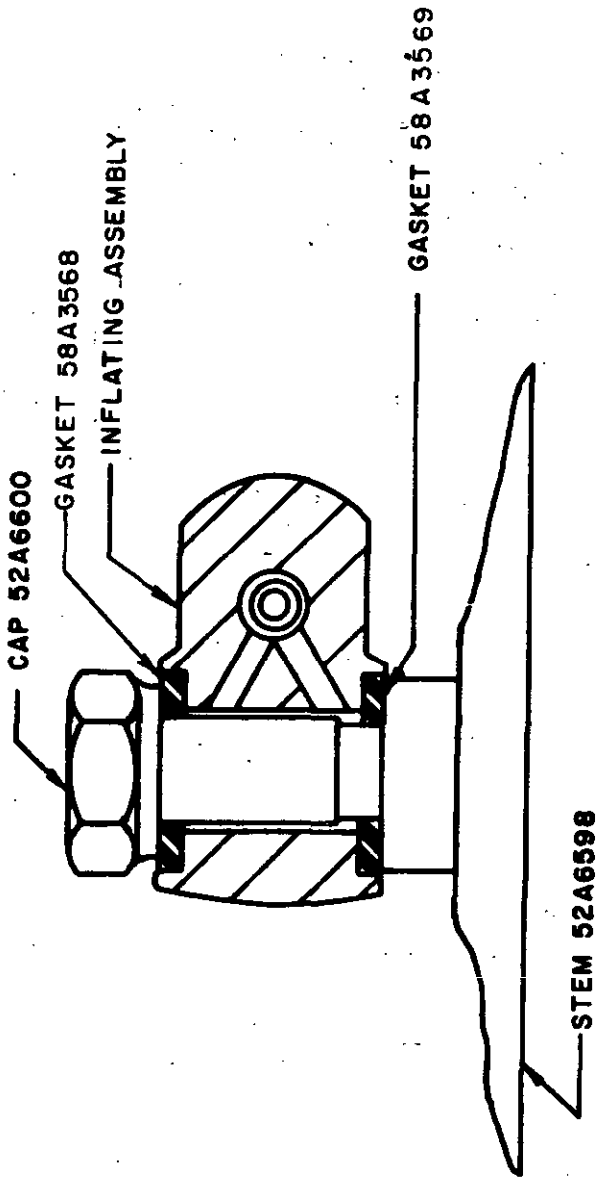


FIGURE 2. Mounting of Inflating Assembly

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discharge a cylinder conforming to MIL-C-25369 in not more than 2-1/2 seconds.

3.6.8 Lock. The lock shall be such that a cylinder installed with a torque of 60-inch pounds can be loosened with a torque of not less than 30 nor more than 60 inch-pounds.

3.7 Identification of product. The inflating assembly shall be marked for identification in accordance with the following specific information as defined in MIL-STD-130:

LIFE PRESERVER INFLATING ASSEMBLY MA-1

MIL-L-25370D(USAF)

Manufacturer's Identification*

Contract or Order No.*

Date of Manufacturer*

*The manufacturer shall insert the applicable information.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Workmanship. The inflating assembly, including all parts and equipment, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship shall be of the highest quality to insure proper functioning of the inflating assembly at all times.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the inflating assembly shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 Test conditions

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, a temperature of 70 (plus or minus 18) degrees Fahrenheit, and a relative humidity of 80 percent or less. If tests are concluded with an atmospheric pressure or temperature different from the starting values, corrections shall be made for the change in values.

4.3.2 Installed cylinders. The cylinders shall be installed with a torque of 60 (plus 10, minus 0) inch-pounds. Discharged cylinders shall be installed in inflating assemblies tested for leakage and hydrostatic pressure.

4.4 Preproduction testing

4.4.1 Test sample. The preproduction test sample shall consist of two inflating assemblies representative of the production equipment. The samples shall be

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identified with the manufacturer's part number and any other information required by the procuring activity (see 6.2). The sample shall be subjected to the preproduction tests specified in 4.4.3.

4.4.2 Test data. Test data shall be in accordance with the instructions of the procuring activity (see 6.2).

4.4.3 Preproduction tests. Preproduction tests shall consist of all tests described under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Individual tests shall consist of the following tests as described under 4.6:

a. Examination of product

b. Operation test without cylinder

c. Lever operation

d. Piercing pin operation.

4.5.2 Sampling tests

4.5.2.1 Lot. A lot shall consist of inflating assemblies manufactured under essentially the same conditions and submitted for inspection at substantially the same time. A lot shall not exceed 500 inflating assemblies for sampling plans A and B and shall be in accordance with inspection level S-3 of MIL-STD-105 for sampling plan C.

4.5.2.2 Sampling plan A. One inflating assembly shall be selected at random from the first lot, and from each fourth lot thereafter, and shall be subjected to the following tests as described under 4.6:

a. Hydrostatic pressure

b. Salt spray.

4.5.2.3 Sampling plan B. Ten inflating assemblies shall be selected at random from each lot of inflating assemblies and shall be subjected to the following tests as described under 4.6:

a. Leakage

b. Operation test with cylinder

c. Lock.

4.5.2.4 Sampling plan C. Upon completion of the individual tests specified in 4.5.1, random samples shall be selected from each lot in accordance with inspection level S-3 of MIL-STD-105. The sample size shall be based only on the applicable size code letter corresponding to the inspection level S-3. Each inflating assembly selected as a sample unit shall be forwarded to an inspection activity selected by the procuring activity (see 6.2) for the following tests as described under 4.6:

a. Examination of product

b. Operation test without cylinder

c. Lever operation

d. Piercing pin operation.

4.5.2.5 Rejection and retest (plans A and B only). When one or more inflating assemblies from a lot selected in accordance with plan A or B fail to meet the specification, acceptance of all inflating assemblies in the lot shall be withheld until the extent

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and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.5.2.6 Rejection (plan C only). Rejection and acceptance for plan C shall be as follows: The acceptance number shall be zero, and the rejection number shall be one for each test. The lot, from which the inflating assembly samples were selected, shall not be shipped until the samples (see 6.3) have been approved by the procuring activity and the lot has been accepted by the cognizant quality control representative. The lot from which the inflating assembly samples were selected shall be rejected if any sample fails to meet any requirement of the tests specified in sampling plan C. A rejected lot shall not be resubmitted except with the approval of the procuring activity.

4.5.2.7 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. However, final acceptance of the entire lot or lots produced later shall not be made until it is determined that all inflating assemblies meet all the requirements of the specification.

4.5.3 Defects in inflating assemblies already accepted. The investigation of a test failure could indicate that defects may exist in inflating assemblies already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. Each inflating assembly shall be visually examined to determine compliance with the requirements of this specification with respect to workmanship, materials, marking, and construction. Each inflating assembly shall be checked to determine if the inlet is threaded to the required depth.

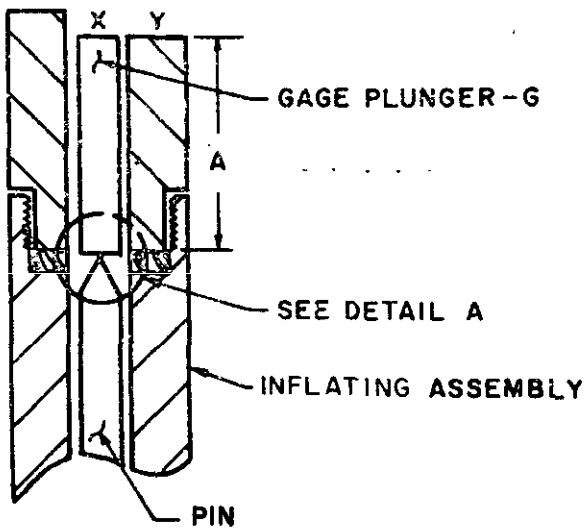
4.6.2 Operation test without cylinder. The inflating assembly shall be held in an inverted position with a 20-pound load attached to the end of the lever arm and with the lever arm held in the normal position. The lever arm shall be released, and the load shall operate the piercing mechanism. The safety thread shall not be used to secure the lever arm during this test. Failure of the inflating assembly to operate the piercing mechanism shall be cause for rejection.

4.6.3 Lever operation. The lever arm of each inflating assembly shall be operated through an arc of not more than 90 degrees. Failure of the piercing pin to pierce the installed cylinder and then to retract to its original position with a travel of the lever through an arc of not more than 90 degrees in line with the longitudinal axis of the inflating assembly shall be cause for rejection.

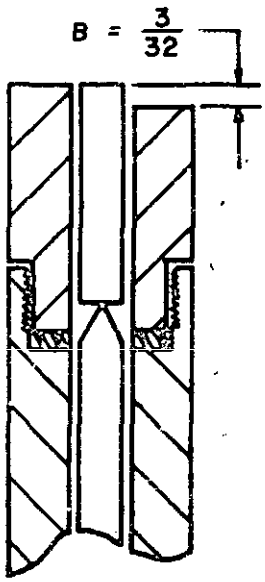
4.6.4 Piercing pin operation. A gage comparable to figure 3 shall be inserted in each inflating assembly to determine if the piercing pin is at least 1/32 inch behind the surface of the inflating assembly gasket (Drawing 58A3567). The lever shall be operated through an arc of 90 degrees; the piercing pin shall travel not less than 3/32 inch beyond the gasket. Surface "X" shall not extend beyond surface "Y" in the unoperated position.

4.6.5 Hydrostatic pressure. A hydrostatic pressure of 1,500 psi shall be applied to the outlet of the inflating assembly for

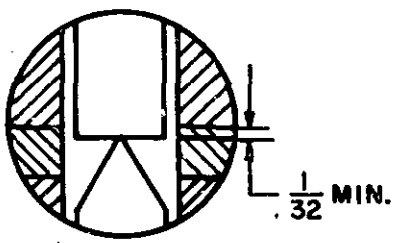
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UNOPERATED POSITION



OPERATED POSITION



DETAIL A

G IS 1/32 INCH LONGER THAN A.
WHEN X IS EVEN WITH SURFACE Y, PIN IS 1/32 INCH BEHIND SURFACE OF GASKET.
B = DISTANCE PIN SHOULD MOVE G WHEN LEVER OF INFLATING ASSEMBLY IS ACUTATED (SEE 4.6.4).
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES AND GAGE DIMENSIONS ARE OPTIONAL.

FIGURE 3. Gage

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a period of not less than 5 minutes. The lever shall be in the operated position. Any deformation or leakage of the inflating assembly shall be cause for rejection. After completion of this test, the inflating assembly shall be subjected to the tests specified in 4.6.6.

4.6.6 Salt spray. The inflating assembly shall be subjected to a 96-hour salt spray test in accordance with method 811 of FED-STD-151. After the salt-spray test there shall be no corrosion of the base metals. After the completion of the salt-spray test, the inflating assembly (with a cylinder conforming to MIL-C-25369 installed) shall be held in an inverted position with a 20-pound load attached to the end of the lever arm and with the lever arm held in the normal position. When the lever arm is released, the load shall operate the piercing mechanism which will pierce and discharge the cylinder in not more than 2-1/2 seconds. The safety thread shall not be used to secure the lever during this test. Failure of the inflating assembly to puncture the cylinder with a force of 20 pounds in 2-1/2 seconds shall be cause for rejection.

4.6.7 Leakage. With the inflating assembly mounted on a valve stem conforming to Drawing 52A6598 with the gaskets and the valve stem cap in place, the lever shall be moved slowly through a normal cycle of operation while the inflating assembly is submerged in water and is subjected to internal pressure of 2 psi for not less than 10 seconds. This test shall be repeated except that the internal pressure applied shall be 40 psi. Any evidence of leakage of the inflating assembly shall be cause for rejection.

4.6.8 Operation tests with cylinder. The inflating assembly shall be mounted on a valve stem conforming to Drawing 52A6598 with gaskets and valve stem cap in place. A cylinder conforming to MIL-C-25369 shall be installed. The inflating assembly shall be held in an inverted position with a load attached to the end of the lever arm and the lever arm held in the normal position. The outlet shall be equipped with a chamber that will provide between 1.5 and 2.5 psi back pressure on the inflating assembly. The inflating assembly and load shall be submerged in water. The load shall be such that a force of 20 pounds is applied to the lever arm. When the lever arm is released, the load shall operate the piercing mechanism which shall pierce and discharge the cylinder in not more than 2-1/2 seconds. The safety thread shall not be used to secure the lever in this test. During and not less than 10 seconds after the lever arm is released, no bubbles shall be visible.

4.6.9 Lock. The lock on a cylinder conforming to MIL-C-25369 installed with a torque of 60 inch-pounds shall be tested by applying a torque, in a counterclockwise direction, to loosen the installed cylinder. Failure of the cylinder to be released with a torque of not less than 30 nor more than 60 inch-pounds shall be cause for rejection of the inflating assembly.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Preservation and packaging for level A shall be in accordance with method III of MIL-P-116. The quantity per unit package shall be one. The inflating assembly shall be packaged in unit containers that conform to PPP-B-566 or PPP-B-676 and that are of minimum size consistent with the

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protection required. Cushioning material conforming to PPP-C-843 shall be applied to the item as necessary to prevent free movement of the item within the unit container.

5.1.2 Level C. Preservation and packaging for level C shall be the minimum necessary to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. *Inflating assemblies* shall be packed in overseas-type containers conforming to PPP-B-576, PPP-B-585, PPP-B-591, PPP-B-636, or PPP-B-640. Containers shall be of minimum weight and cube, uniform size and shape, and when practical, shall contain like quantities. The gross weight of the containers shall be governed by the applicable container specification. Closure of the containers shall be in accordance with the applicable specification and appendix thereto. When containers conforming to PPP-B-636 are used, all joints and seams shall be sealed with tape conforming to PPP-T-60.

5.2.2 Level B. For level B packing inflating assemblies shall be packed as specified in 5.2.1, except that the containers shall be of the domestic type and closure shall be in accordance with the requirements for domestic containers.

5.2.3 Level C. Packages requiring overpacking for acceptance by the carrier shall be packed in shipping containers in a manner that will ensure safe delivery at the lowest transportation and packing charges. Containers shall meet Consolidated Freight Rules or any other rules and regulations applicable to the mode of transportation.

5.3 Marking. Unit, intermediate, and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The inflating assembly covered by this specification is intended for use as a component of the Types MB-1, MD-1, and LPU-2/P life preservers.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Preproduction requirements (see 4.4.1)

c. Requests for results of preproduction tests (see 4.4.2)

d. Selection of inspection activity for sampling plan C (see 4.5.2.4)

e. Selection of applicable levels (see 5.1 and 5.2).

6.3 Test samples. The test samples specified in 4.5.2.6 should be returned to the manufacturer.

6.4 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Preparing activity:
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Reviewers:
Air Force - 11, 84

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